

Brian Lembuss

Aerospace Engineer | Technical Founder | Deep-Tech Researcher | Academic Educator

Nairobi, Kenya | brianlembuss@gmail.com | +254 705 875103 | <https://www.linkedin.com/in/brian-kipchumba-lembuss-kirwa/>
| <https://github.com/lembuss/>

PROFESSIONAL PROFILE

Aerospace Systems Engineer, Technical Founder and Deep-Tech Researcher with six years of progressive experience spanning manned aircraft maintenance, UAV systems engineering, eVTOL flight test management, cloud software architecture, and university instruction. Holds a B.Sc. in Aerospace Engineering with honours from METU and 73 ECTS toward an M.Sc. in Aerospace at TU Munich (degree completion pending). Career includes commercial MRO on turboprop/turbofan aircraft (ALS Ltd); GCS serial production and server-rack HIL test rig builds for commercial UAVs (Amazilia Aerospace); aerodynamic design loop engineering and successful transition flight testing on a 25kg MTOW eVTOL (HORYZN); published airspace throughput research (AIAA 2024); flight simulator instruction (TUM); and founding leadership of Kipepeo Aerospace covering platform CAD, embedded C++ software, GCP cloud microservices, and SAFE equity fundraising. Brings experience across the hardware/software boundary, combining physical aircraft systems lifecycle ownership with cloud, AI, and embedded software architectures.

CORE COMPETENCIES

Aircraft Systems & Aerospace Engineering: Systems engineering lifecycle, Requirements engineering, Functional decomposition, System architecture, Interface definition, Hardware/software integration, Avionics, Flight controls, Embedded systems, HIL testing, Verification & validation, Flight testing, MRO, UAS, eVTOL, Rotorcraft, Fixed-wing aircraft, Aircraft design, Ground control systems, MBSE

Venture Building & Entrepreneurial Leadership: Technical venture creation, Company incorporation, Corporate governance, Cap Table structuring, Fundraising, SAFE financing, Grant acquisition, Commercial partnerships, MoU negotiation, Accelerator programmes, Market discovery, Product development, Technical leadership, Recruitment

Deep-Tech Research & Computational Capability: Applied aerospace research, Computational modelling, UAM simulation, Aerodynamic CFD, Embedded C++, Test-driven development, Autonomous systems, ROS, 3D perception, MBSE, AI/ML, Cloud systems, Data pipelines, Photogrammetry, OpenFOAM, ANSYS CFX, MATLAB, Python, C++, TypeScript/Node.js/React

Teaching, Mentorship & Academic Service: University instruction, Flight simulator instruction, Examining, Numerical methods, SI-PASS tutoring, STEM mentorship, Curriculum support, Academic administration, Student organisation leadership

PROFESSIONAL EXPERIENCE & SELECTED R&D; PROJECTS

Founding CEO & Lead Systems Engineer — Kipepeo Aerospace (Startup)

Nairobi, Kenya | Oct 2024 – Present

Founding CEO and Lead Systems Engineer driving the full-stack design, hardware fabrication, software architecture, and commercial strategy for autonomous UAV platforms, multispectral payload hardware, cloud photogrammetry pipelines, and AI intelligence systems across precision agriculture and defense verticals in East Africa.

TAI UAS - Tactical Aerial Intelligence Platform (R&D)

- Executed the initial aircraft design lifecycle for the TAI UAS, capturing system-level requirements and conducting preliminary aerodynamic and sizing trade studies for a 2-meter delta-wing hybrid VTOL platform
- Authored the complete system architecture for the TAI UAS, translating high-level mission requirements into functional allocations, subsystem definitions, and a detailed electrical/control schematic
- Modeled the 3D mechanical CAD assembly for the TAI delta-wing hybrid VTOL airframe in Autodesk Inventor, incorporating internal structural housing for avionics, propulsion mounts, and modular payload bays
- Conducted market trade studies, component matching, and cost estimations to compile the full Bill of Materials (BOM) for the TAI UAS platform

Kilimo Anga — Quadrotor Testbed & AngaCam Payload Integration

- Derived technical specifications and operational requirements for an agile, low-cost quadrotor testbed and multispectral payload based on farmer feedback from the Kilimo Anga initiative.
- Engineered the electrical, power distribution, and flight control architecture for the F450-based quadrotor testbed.

- Designed the 3D mechanical housing and optical mounting for the custom AngaCam multispectral payload in Autodesk Inventor.
- Performed component trade studies and compiled the master Bill of Materials (BOM) for the quadrotor testbed and AngaCam payload.

AngaStack V1 Platform Development (R&D)

- Local Photogrammetry Pipeline Development & Algorithm Verification
- Full-Stack Application Development (AngaView UI & AngaCloud Backend)
- Azure Cloud Infrastructure Provisioning & Live Platform Deployment
- AngaDemo Digital Prototype Build & Field Usability Validation

Kilimo Anga - Field Operations, Pre-Pilot Readiness & Business Model

- Farmer Persona Discovery
- Competitor Gap Analysis & Fail-Point Mapping
- Cost of Status Quo (CoSQ) Financial & Impact Modeling
- Credit-in-Kind Business Model & Value Chain Offloading Design
- Pre-Pilot Ecosystem Partnerships, Governance & Investor Readiness

AngaStack V2 Platform Development - GCP Cloud Migration, Mobile Ingestion & AngAi Engine (R&D)

- GCP Infrastructure Provisioning & IAM Security Architecture
- Photogrammetry Pipeline GCP Migration & Numerical Stats Extraction
- Backend Service Refactoring, Dual-Path Ingestion & Job Orchestration
- AngAi Agronomic Reasoning Engine & RAG Microservice Build
- AngaView Mobile Ingestion, Firebase Hosting & Living Farm Health Record UI

Linda Nchi — Sovereign Tactical ISR Platform & K-DEMO-2.5 Cargo Demonstrator (R&D)

- Defense Stakeholder Requirement Synthesis & Use-Case Mapping
- Linda Nchi Tactical ISR System Architecture & Technical Roadmap Authoring
- Project K-DEMO-2.5 Heavy-Lift Cargo Hexacopter Mechanical Engineering & Stress Analysis

Kipepeo Venture Building, Governance & Capital Strategy

- Formulated the foundational vision, operational roadmap, and financial budget for establishing an indigenous drone technology hub at Konza Technopolis
- Executed formal business name registration and statutory setup to legally establish Kipepeo Aerospace in Nairobi, Kenya
- Participated in the 3-month Jasiri Talent Investor residential intensive in Bugesera, Rwanda, conceptualizing the iCARUS precision agriculture venture and conducting smallholder agri-data market research
- Applied for and secured non-dilutive cloud infrastructure grant funding through the Microsoft for Startups Founders Hub program
- Negotiated and executed three strategic Memorandums of Understanding (MoUs) with Stablegen Investment Ltd. to establish pilot market access, software testing, and drone corridor access
- Registered core internet domains, established digital brand presence, and deployed web infrastructure for Kipepeo Aerospace and its flagship initiative Kilimo Anga
- Conducted talent acquisition campaigns, restructured executive roles, and established a specialized advisory board to strengthen Kipepeo Aerospace's commercial, legal, and agronomic leadership
- Applied for the MK-Africa #MyLittleBigThing Sustainable Venture Challenge with Kilimo Anga, securing selection into the Top 50 and completing the two-day Innovation Bootcamp.
- Executed the 30-Day Build & Test Challenge for #MyLittleBigThing, deploying the AngaDemo digital prototype and conducting field usability and willingness-to-pay (WTP) validation studies
- Applied for and secured non-dilutive cloud infrastructure grant funding from the Google Cloud for Startups program
- Completed the Viktoria Ventures Startup360 Access to Finance (A2F) Programme and authored the \$30,000 pre-seed investor readiness dossier and financial model for Kilimo Anga
- Authored and executed the binding Kipepeo Aerospace Pre-Incorporation Founders' Agreement, establishing legal governance, cap table ownership, reverse vesting, and operational decision hierarchies under Kenyan law
- Negotiated and executed the first formal early-stage investment agreement for Kipepeo Aerospace via a Simple Agreement for Future Equity (SAFE) with Zimbu Investments Limited.
- Applied for and onboarded into the 12-Month #MyLittleBigThing Investor Readiness Program (IRP) Pilot Cohort (2026–2027) co-delivered by MK-Africa and Strathmore @iBizAfrica

Jasiri Fellow, Talent Investor Program — Jasiri4Africa (Entrepreneur)

Rwanda | Jan 2025 – May 2025

Participated in the Jasiri Talent Investor program (Cohort 7 / Ntore Cohort) funded by Allan & Gill Gray Philanthropy. Completed the online Jumpstart phase and 3-month Residential Intensive in Bugesera, Rwanda, focusing on team formation, qualitative/quantitative smallholder farmer discovery, commissioning a GeoPoll research study, formulating the iCARUS business model canvas, and presenting at Jasiri Demo Day 1 before opting out prior to the venture creation phase.

Jasiri Fellow, Talent Investor Program

- Completed the Jasiri Jumpstart and 3-month Residential Intensive while executing field-level market discovery on smallholder agri-data adoption barriers in Kenya.
- formulated the dual-sided business model canvas and revenue architecture for the iCARUS precision agriculture concept.
- Authored and delivered the iCARUS venture pitch deck at Jasiri Demo Day 1 in Bugesera, Rwanda

Aerospace Systems Engineering — Amazilia Aerospace GmbH (Working Student)

Munich, Germany | Sep 2022 – Dec 2024

Role concurrent with MSc studies at TU Munich. Involved in aircraft systems integration, electrical architecture, HIL testing, GCS development and autonomous charging systems across both UAS and conventional aircraft platforms.

WfA MiniFreighter GCS -SN2 (R&D)

- Attended joint design evaluation session with Wings for Aid CEO and chief pilot alongside Amazilia CTO and senior systems engineer. Captured customer and pilot requirements directly and translated them into formal engineering specifications for the SN2 GCS redesign. Produced wiring diagrams, CAD renders and system architecture documentation. Presented across three internal design reviews with CTO, Head of Systems Engineering and Senior Systems Engineer, incorporating embedded software team input on DCU interface changes. Received sign-off to proceed to BOM and procurement.
- Sourced alternative panel supplier through independent research, which reduced component costs compared to the original source. Redesigned front panels in Autodesk Inventor to match the new supplier's specifications. Curated a full assembly procedure based on SN1 documentation and discussions with seniors. Upon parts arrival, performed mechanical assembly including drilling connection holes on panels and GCS case, installing internal support structures, and routing the power architecture — standardising all internal components on 12VDC via an AC/DC converter.
- Performed stepwise electrical integration of all internal GCS components following the assembly procedure, conducting QA checks and functionality tests at marked checkpoints verified with seniors. Installed GCS software received from the software team upon completion of Unit 1. Scheduled and conducted simulator testing in the lab to evaluate control software, internal wiring of controls, stick feedback and aircraft response time. Applied personal drone and private pilot experience to assess pilot feel and control responsiveness directly during simulation runs. Coordinated firmware and wiring corrections on the hardware side while communicating required software changes to the software team. Conducted a final internal review before inviting the Wings for Aid CEO and chief pilot for customer acceptance testing against the original specifications.
- Serially produced four additional GCS units following customer acceptance of Unit 1. Worked on all four units simultaneously in a dedicated assembly space, applying the verified assembly procedure and QA process established during Unit 1 production. Each unit went through the same integration, functionality testing and verification process before being cleared for delivery

Amazilia Ground Control Station (R&D)

- Reviewed and cleaned up existing CAD files, wiring diagrams and system architecture documentation inherited from predecessor. Reconciled documentation against the physical design to ensure accuracy before proceeding to assembly.
- Performed mechanical assembly of three tabletop GCS units including mounting of three-monitor arrangement, mode control panel and internal structural components. Drilled connection points and installed internal support elements following the inherited design
- Performed electrical integration of all internal components across three units including power architecture, monitor connections, mode control panel wiring and communication interfaces. Followed stepwise approach with QA checks at key points.
- Conducted functional testing of each completed unit, verifying correct operation of all three monitor feeds — camera, telemetry and engine parameters — as well as mode control panel inputs and communication interfaces. Validated that each unit performed correctly within the lab environment across the aircraft types Amazilia was working on — Pipistrel OPV, Nuuva and CTOL.

Aircraft Systems Hardware-in-the-Loop (HIL) Test Rig (R&D)

- Received system requirements from Head of Systems Engineering defining the HIL rig scope, component specifications and communications architecture. Studied the complex multi-protocol communications architecture of the aircraft systems. Developed the high-level wiring architecture covering OPS Panel and PSUs, CAN boxes, RPI and USB hub, FCCs and

DCUs, ETH switch, LMUs, Panels A/B/C and BRS tray. Produced detailed wiring sheets and interface definitions across all components.

- Designed the full server rack assembly in Autodesk Inventor, modelling the physical layout and component placement of the HIL rig including rack structure, trays, panels and internal component arrangement. Presented the design across multiple preliminary design reviews with cross-functional teams including systems engineering and embedded software before receiving sign-off to proceed to procurement and assembly
- Procured all components for the HIL rig following design sign-off. Performed mechanical assembly of the server rack including mounting of rack rails, trays, panels and structural elements. Installed and positioned all hardware components within the rack according to the approved CAD layout
- Performed full electrical integration of all HIL rig components including power distribution, CAN bus wiring, Ethernet patch panel connections, and interface wiring between FCCs, DCUs, LMUs, RPi units and OPS panels. Followed the wiring architecture developed during design, working stepwise through each subsystem layer of the rack.
- Conducted extensive testing and verification of the completed HIL rig with seniors, validating correct operation of all interfaces, communication protocols and hardware components. Progressed through to full HIL environment testing — monitoring and controlling aircraft systems in simulated flight conditions from the lab. Verified that the rig faithfully replicated the actual aircraft avionics architecture and communication behaviour.

Aircraft Battery Charging Unit (ABCU) (R&D)

- Following a polarity reversal incident during field testing of a Pipistrel aircraft that caused a short circuit and wire damage, received brief from CTO to design a safe standalone aircraft battery charging unit. Conducted requirements capture by consulting all field testing personnel to understand operational needs, environmental conditions and safety requirements. Researched LiFePO₄ battery chemistry, charging characteristics and available charging system suppliers to establish a technically grounded requirements baseline.
- Developed system architecture for the ABCU covering power input, charging system selection, voltage switching between 12V and 24V LiFePO₄ configurations, polarity protection, battery voltage indication and enclosure design. Conducted detailed component market study matching available charging systems against Amazilia's battery specifications. Selected components capable of handling both 12V and 24V LiFePO₄ chemistries with switchable output, allowing field teams to switch between configurations during operations. Optimised component selection for cost following design review feedback.
- Produced full wiring diagrams for the ABCU internal architecture. Designed enclosure panels, panel holders, mounting hardware and overall box layout in Autodesk Inventor, sized for transport in a hardened carry case similar to those used for the GCS units. Designed front panel layout for voltage switching and indicators. Specified screw sizes, mounting arrangements and all mechanical details required for assembly. Presented complete design package across design reviews with CTO and senior systems engineering team, incorporating revisions to optimise cost and usability. Agreed final panel supplier — same alternative supplier used for the WfA GCS — following design review approval.
- Upon departure from Amazilia, compiled and handed over the complete ABCU design package to the team. Handover included full system architecture diagrams, wiring diagrams, component datasheets and specifications, BOM, panel CAD designs and a step-by-step assembly guide to enable the team to proceed to physical build without loss of design intent or technical context

Project Management - Systems & Integration — HORYZN (Student Initiative)

Munich, Germany | Jul 2023 – Jul 2024

Appointed Project Manager for systems integration and flight testing phase of the Kolibri hybrid lift-and-cruise eVTOL UAS under Mission Pulse Phase 2. Oversaw cross-functional coordination across avionics, flight testing, CAD and structures teams. Hands-on involvement in avionics integration, PX4 flight controller setup, ground testing, flight testing and regulatory preparation. Secured TU Munich aerospace test facilities at Oberpfaffenhofen for final flight campaign. Project concluded with successful transition flight test — completing the Mission Pulse programme. Concurrent with MSc studies and Amazilia working student role. Held EU A1/A3 drone pilot licence throughout this period.

Kolibri eVTOL — Systems Integration, Flight Testing & Project Completion (R&D)

- Appointed Project Manager for systems integration and flight testing phase of Kolibri by the outgoing project management team, based on demonstrated technical contribution and cross-team familiarity during the aerodynamics phase. Transitioned from aerodynamics engineering role into project leadership, taking responsibility for coordinating the avionics and flight testing team as primary focus, with secondary oversight of CAD and structures teams. Obtained EU A1/A3 drone pilot licence around this time, enabling direct participation in flight operations.

- Served as the central coordination point between avionics, flight testing, CAD and structures teams during the systems integration phase of Kolibri. Identified and resolved integration conflicts — for example, where avionics and electronics sizing did not fit within the structural spaces allocated by the CAD team, requiring coordinated design iterations across CAD, structures and avionics teams. Facilitated regular cross-team meetings, tracked outstanding integration issues and ensured design decisions were communicated and implemented consistently across all teams. Coordinated with other project managers on resource allocation and scheduling.
- Worked hands-on with the avionics and flight testing team on the integration of Kolibri's avionics systems including flight controller setup on PX4, ESC configuration, sensor integration and onboard electronics wiring. Applied systems integration experience from Amazilia to support the team in solving avionics interface and communication issues. Contributed directly to soldering, wiring and bench testing of avionics components. Used knowledge of Kolibri's full design — aerodynamics, structures and avionics — to provide informed technical guidance on integration decisions.
- Participated in ground testing and hover trial campaign for Kolibri. Built and operated a hover test bench to validate lift motor thrust ratios against the design requirements for the 25kg MTOW aircraft. Tested the defibrillator payload delivery system to confirm correct operation with the aircraft systems. Analysed flight logs from test runs to identify issues and drive corrective actions. Managed the iterative build-test-fail-repeat development cycle — including analysis of two crashes — using data from flight logs and ground control interfaces to diagnose faults and implement fixes.
- Supported preparation of regulatory documentation for Kolibri flight operations under the SORA framework. Coordinated with the team to ensure operational risk classifications and flight test documentation met the requirements for test flights at the secured facility. Contributed to safety planning and operational procedures for the flight test campaign.
- Secured access to a TU Munich aerospace test facility for the Kolibri flight test campaign, coordinating with university contacts to establish a dedicated workspace with appropriate flight test area. Managed all logistics for the flight campaign including team transport, equipment movement, scheduling and on-site coordination. Maintained sponsor communications throughout, providing updates on project progress and test milestones. Coordinated new member recruitment and interviews for incoming semester intake alongside other project managers.
- Participated in the final Kolibri transition flight test campaign at Oberpfaffenhofen test airport. Contributed to pre-flight preparation, on-site coordination and post-flight analysis. Kolibri successfully completed VTOL hover, transition to forward flight and cruise phases — achieving the core Mission Pulse objective that the predecessor prototype Frankenstein had failed to reach. Mission Pulse Phase 2 concluded successfully with the completion of the transition flight test.

Aerodynamics Project Engineer — HORYZN (Student Initiative)

Munich, Germany | Oct 2022 – Jun 2023

Unpaid student initiative at TU Munich. Member of the Design Loop team responsible for the aerodynamics module of the Kolibri hybrid lift-and-cruise eVTOL UAS under Mission Pulse Phase 2. Led the aerodynamics sub-team of three engineers. Developed and maintained the aerodynamics module within a CPACS/MATLAB/PAWAT/RCE integrated design environment, contributing to aircraft configuration selection and iterative multidisciplinary design optimization concurrent with MSc studies and Amazilia working student role.

Kolibri eVTOL — Aerodynamics Module Development (R&D)

- Participated in structured handover sessions with the outgoing Mission Pulse Phase 1 team to understand the design, shortcomings and failure modes of the Frankenstein prototype. Analysed the previous CPACS files, MATLAB code and PAWAT outputs to reconstruct what had been built and why transition had failed. Collaborated across the full Design Loop team to document lessons learned and define requirements list for the Phase 2 prototype — Kolibri. Requirements captured and managed in Notion.
- Led the aerodynamics team's contribution to the Kolibri configuration selection process. Identified the root cause of Frankenstein's transition failure as wingtip-mounted propellers creating adverse interference at the 25kg MTOW scale. Proposed a Cessna-type fixed-wing configuration with a front tractor propeller and lift motors mounted on the wing, supported by case studies from undergraduate aircraft design projects and CFD analysis in OpenFOAM demonstrating adequate glide performance during transition in the event of propulsion switching delays. Presented the configuration argument across multiple design reviews, convincing the team and securing agreement to proceed.
- Developed and maintained the aerodynamics module within the HORYZN multidisciplinary design loop. The module interfaced with CPACS via TiGL and TiXi libraries, using PAWAT as the core aerodynamic solver within a MATLAB class-based architecture. Implemented methods covering airfoil database management, CPACS geometry parsing, aerodynamic mapping across Reynolds number and angle of attack ranges, twist optimisation, stall velocity and force calculations, neutral point calculation and results writing back to CPACS. Wrote full module documentation including README, UML class diagram, function descriptions and I/O specifications. Managed module codebase on Git.
- Participated in the aerodynamics team's airfoil selection process for Kolibri's lifting surfaces. Ran systematic case studies across candidate airfoils using XFLR5 and Flow5 to evaluate CL/CD performance across the expected Reynolds number range. Fed selected airfoil profiles into the CPACS aircraft definition and ran iterative optimisation of wing geometry including twist distribution, span, chord and sweep through the aerodynamics module. Coordinated with structures and

geometry sub-teams within the design loop to ensure aerodynamic outputs were compatible with structural and sizing constraints.

- Worked intensively within the RCE-based multidisciplinary design loop alongside aerodynamics, structures, geometry, wing sizing and propulsion sub-teams to achieve design convergence for Kolibri. Ensured aerodynamics module outputs were compatible with and correctly consumed by adjacent modules. Participated in extended working sessions — frequently 18 to 20 hours — to resolve inter-module conflicts and iterate toward a stable baseline design. Contributed aerodynamics expertise to cross-team configuration discussions and design decisions throughout the convergence process
- Prepared and delivered a comprehensive handover of the aerodynamics module to the incoming team and downstream design teams. Handover package included the full codebase on Git, README documentation, UML class diagram, function descriptions, I/O path specifications, calculation options documentation, Flow5 validation task documentation and a formal handover document. Ensured the receiving team could understand, operate and extend the module without loss of design intent or technical context.

UAS Engineer, Pilot and Instructor — Kendrone Ltd (UAS Company)

Mtwapa, Nairobi, Naivasha - Kenya | Jan 2021 – Jun 2021

Joined Kenya's first KCAA-approved Drone Training Organisation as a student, earning the Remote Pilot Licence (YK-RPL-00013A) before transitioning into a permanent employee role as UAS Pilot, Instructor, and IT Assistant. Delivered ground school and practical flight instruction across three student cohorts, completed the Instructor Rating (IR001-01) under a company-sponsored training bond, and led engineering and innovation work including the design, fabrication, and flight testing of a drone-based seedball dispersal mechanism for reforestation in collaboration with Seedball Kenya. Concurrently owned all IT infrastructure, digital marketing, student administration, and external stakeholder engagement for the company.

Drone-Based Seedball Dispersal Mechanism (R&D)

- Captured mission requirements from CEO and Seedball Kenya and produced mechanical design of seedball container and servo-actuated release mechanism
- Fabricated mechanism components and integrated the dispersal payload onto the Tarot 650 airframe including servo power rail decoupling
- Prepared Tarot 650 platform for flight operations — diagnosed and resolved flight controller calibration and motor imbalance issues, configured failsafes, and conducted initial airworthiness verification flight
- Executed 3-flight endurance test campaign to characterise platform performance under operational payload conditions and authored formal test report
- Conducted witnessed operational demonstration of the seedball dispersal system before Seedball Kenya, KCAA-adjacent observer, and company CEO

UAS Pilot Training Programme Delivery

- Delivered a 32-slide technical presentation on BVLOS UAS operations to a KCAA representative, chief instructor, and CEO as part of the instructor rating certification process
- Delivered full KCAA RPL ground school theory programme across three student cohorts covering seven subject domains
- Delivered 15-lesson KCAA RPL practical flight instruction programme across three student cohorts on multirotor platforms

Aerial Mapping & Agricultural Survey Operations

- Executed end-to-end aerial survey and agricultural mapping missions as PIC, from mission planning through photogrammetric processing to final deliverable outputs
- Trained professional surveyors from external survey companies in UAS-based mapping workflows during live operational missions

UAS Avionics, Payload Integration & Fleet Maintenance

- Integrated avionics and sensor payloads across the Kendrone multirotor fleet for mapping and agricultural missions
- Diagnosed and performed component-level hardware repairs on DJI multirotor platforms to restore airworthiness
- Conducted systematic pre- and post-flight airworthiness inspection routines across the Kendrone fleet

IT Infrastructure, Digital Presence & Company Operations

- Diagnosed and resolved company email deliverability failures and optimised web infrastructure
- Reorganised company and student databases and maintained operational records including petty cash
- Designed company marketing collateral and managed digital marketing channels including paid advertising
- Led external stakeholder engagement across regulatory bodies, media, landowners, aviation doctors, and commercial clients

Aircraft Maintenance Intern — ALS Ltd (Internship)

Wilson Airport, Nairobi | Jul 2017 – Sep 2017

Hands-on MRO internship at ALS Ltd, a KCAA-certified aviation company based at Wilson Airport Nairobi, conducted as a mandatory industrial attachment for the BSc Aerospace Engineering programme at METU. Worked under the supervision of licensed engineers in the Base Maintenance department, assisting with 200HR inspections, line maintenance, and C-check tasks across a mixed turboprop and turbofan fleet including the Beechcraft 1900C (PT6A-65B), De Havilland DHC-8 (PW121), and Embraer ERJ145MP (Rolls-Royce AE 3007A). Supervised by Base Maintenance Manager Maurice Mukung

Aircraft Line, Base & C-Check Maintenance — ALS Ltd MRO

- Assisted with fuel nozzle replacement on the PT6A-65B turboprop engine during a 200HR inspection on the Beechcraft 1900C
- Assisted with compressor wash on the PT6A-65B turboprop engine during a 200HR inspection on the Beechcraft 1900C
- Assisted with propeller shaft greasing on the PT6A-65B turboprop engine during a 200HR inspection on the Beechcraft 1900C
- Assisted with de-icer boot reinstallation on the DHC-8 leading edges during line maintenance
- Assisted with flap ball screw assembly lubrication on the DHC-8 during line maintenance
- Assisted with cabin seat removal and hook-and-loop fastener replacement on the Embraer ERJ145MP during C-check
- Assisted with nose landing gear servicing on the Embraer ERJ145MP during C-check

ADDITIONAL RESEARCH & ACADEMIC PROJECTS

Generic Modeling of Slotneutral UAM Throughput at Commercial Airports (R&D) — Technical University of Munich

2021 – Present

- Traffic Categorization & Glideslope Separation Analysis
- PinS Approach Procedure & Holding Pattern Trajectory Design
- Python Throughput Modeling Toolchain & KML Trajectory Generation
- AIAA 2024 Research Contribution — Airspace Geometry Dynamization

Embedded Software Development of the Actuator Control and Monitoring Unit (ACMU) (R&D) — Technical University of Munich

2021 – Present

- Refactored the ACMU data logging software pipeline using Test-Driven Development (TDD) in C++ to implement high-efficiency binary data serialization over USB
- ACMU System Architecture & Protocol Integration Research

Autonomous Sub-Terrain UAV Challenge (R&D) — Technical University of Munich

2021 – Present

- Depth-to-PointCloud Perception Engine & Spatial Frame Transformation
- OctoMap 3D Voxel Grid Mapping & Dynamic Point Cloud Updater

Applied CFD Channel & Cavity Flow Simulation (R&D) — Technical University of Munich

2021 – Present

- Modeled 2D/3D dual-cavity channel geometries, built multi-block structured grids, and executed comparative subsonic CFD simulations across 5 test cases in ANSYS ICEM CFD and ANSYS CFX

Simulation of the Flowfield of a Multirotor in Axial Flight (R&D) — Technical University of Munich

2021 – Present

- OpenFOAM Multirotor CFD Mesh Generation & Parallel Domain Setup
- BEMT Actuator Disk Source Integration & Axial Flow Field Simulation

Geospatial Analysis & Cartography — Nakuru County Geotourism Research (R&D) — Technical University of Munich

2021 – Present

- Formulated the spatial data processing pipeline and generated all regional GIS figures, geosite localizations, and geological mapping overlays.

Design and Implementation of an Unmanned Aircraft System (UAS) using the Model Based System Engineering (MBSE) Approach for Agricultural Applications at the Galana Kulalu Irrigation Scheme (R&D) — University of Cologne

2022 – 2023

- UAS system architecture definition and MBSE modeling for autonomous precision agriculture operations

- Precision agriculture capability definition — crop health monitoring, soil fertility and water retention analysis, and chemical spraying payload architecture
- Work breakdown structure, critical path, and Gantt chart development for UAS implementation at Galana Kulalu
- SWOT and PEST analysis — individual strengths, weaknesses, opportunities, threats, and external risk and opportunity assessment for the UAS project
- Project summary write-up and elevator pitch preparation and delivery

Numerical Methods for Aerospace Engineers — Middle East Technical University

2015 – 2019

- Developed a 2D finite volume solver in FORTRAN to compute potential flow over a NACA 0012 airfoil and a circle at multiple angles of attack.
- Implemented and compared explicit and implicit 1D finite difference schemes for an unsteady heat diffusion equation (parabolic PDE) and a linear convection equation (hyperbolic PDE).
- Implemented FTCS and DuFort-Frankel schemes to solve the 2D unsteady convection-diffusion equation over a rectangular domain, with parametric studies on grid resolution, convection velocity, and initial condition geometry

Finite Element Modelling of a Composite Cantilever Beam — Middle East Technical University

2015 – 2019

- Performed static 3D FEA of a 5-layer composite cantilever beam (3 aluminium + 2 glass fibre/epoxy, 500mm × 50mm) under a 5N transverse tip load to determine tip deflection and full stress distribution across all layers.
- Performed multi-analysis 3D FEA of a 3-layer composite cantilever beam (1 aluminium core + 2 glass fibre/epoxy, 200mm × 40mm) under static and harmonic torsional moment loading ($M_x = 1\text{Nm}$), including modal frequency extraction and harmonic response analysis.

Turbofan Compressor Performance Analysis (R&D) — Middle East Technical University

2015 – 2019

- Conducted parametric analysis of turbofan compressor performance by modelling the effects of turbine inlet temperature, operating altitude, and turbine inlet area on corrected mass flow rate and compressor pressure ratio

Finite Difference Method Solvers for Aerodynamic PDEs — Middle East Technical University

2015 – 2019

- Led development of four MATLAB finite difference solvers covering elliptic, parabolic, and hyperbolic PDE classes, applying ASE301 numerical methods experience to architect the solution approach and implement stability-verified codes.

Conceptual Design of a Fixed-Wing Luxury Aircraft (Graduation Project I) — Middle East Technical University

2015 – 2019

- Defined customer design requirements and conducted a structured competitor benchmarking study across six business jet aircraft to anchor the conceptual design baseline.
- Estimated initial take-off gross weight W_{TO} using mission segment weight fractions and conducted parametric trade studies to characterise W_{TO} sensitivity to range, endurance, and payload.
- Selected the wing airfoil through comparative aerodynamic analysis of five candidate profiles and sized all primary wing geometry parameters.
- Determined final thrust-to-weight ratio and wing loading through multi-constraint parametric analysis, then refined the take-off gross weight estimate using updated mission segment fractions.
- Sized all major airframe components — fuselage, wing, tail surfaces, and engines — and estimated the aircraft centre of gravity from component weight and moment arm data
- Designed the landing gear configuration, sized tyres and struts, validated geometric placement, and produced the final three-view layout drawing.

Conceptual Design of a VTOL Personal Air Vehicle (Graduation Project II) — Middle East Technical University

2015 – 2019

- Defined customer design requirements and benchmarked six light civil helicopters to establish the design baseline.
- Selected the rotor airfoil and sized the main and tail rotors through Blade Element Momentum Theory analysis across three disk loading cases
- Estimated helicopter component weights iteratively using empirical group equations and selected the engine based on power requirements.
- Computed total power required across the forward flight speed envelope to validate engine selection and characterise helicopter performance.
- Defined all major design configuration choices and produced the final three-view engineering drawing.

Design and Development of an Automated Air Filter — Middle East Technical University

2015 – 2019

- Defined system requirements, selected all electromechanical components, and produced the full circuit schematic.
- Developed Arduino firmware implementing sensor reading, dust density calculation, threshold-based control logic, and all output actuation
- Fabricated the device housing, assembled all subsystems into a unified physical unit, resolved integration faults, and validated full system operation.

Simulation of a piezoelectric d15 torsion sensor (R&D) — Middle East Technical University

2015 – 2019

- Established the theoretical framework for the d₁₅ piezoelectric sensing effect and defined the full PIC255 material model in Abaqus CAE with dual-polarization patch assignment.
- Built the full Abaqus CAE finite element model of the cantilevered piezoelectric beam including geometry, boundary conditions, constraints, and mesh.
- Executed three sequential FEA analyses — quasi-static, modal, and harmonic — to extract sensor voltages and electromechanical coupling coefficients across torsional modes.

EDUCATION

M.Sc Aerospace — Technical University of Munich (Munich, Germany)

Oct 2021 – Present

Status: Degree completion pending (73 ECTS completed; 17 credits and thesis remaining).

LEAD! Leadership for Africa — University of Cologne (Cologne, Germany (Hybrid — online and on-campus))

Feb 2022 – Jan 2023

Status: 260 units / 195 hours. Grade: Sehr gut (A), 8 ECTS.

B.Sc. Aerospace Engineering — Middle East Technical University (Turkey/Northern Cyprus)

Sep 2015 – Jun 2019

Status: Graduation Honors: **B** / Honour Classification (CGPA: 3.12/4.00, 259.5 ECTS).

RESEARCH & PUBLICATIONS

- **Co-Author**, "Conceptualization and simulation of Advanced Air Mobility (AAM) operations within Controlled Airspaces of airports running independent parallel runway operations" — AIAA 2024 Aviation Forum
- **Co-Author**, "Geotourism Potential of Geological Resources in Selected Regions of Nakuru County, along the Gregory Rift, Kenya" — Peer-reviewed journal article

TEACHING & ACADEMIC EXPERIENCE

Student Assistant - IFR Simulator Instructor — Technical University of Munich — Institute for Rotorcraft and Vertical Flight

Munich, Germany | Apr 2023 – Sep 2024

Instructed and evaluated aerospace engineering students during the MW1450 IFR Helicopter Flight Lab course across three academic semesters (Summer 2023, Winter 2023/24, Summer 2024). Conducted 8 weekly 5-hour practical training sessions per semester for student pairs, teaching IFR procedures, emergency protocols, radio telephony, and digital cockpit systems on an EC135 fixed-base simulator. Administered and evaluated 5-hour oral and practical final examinations as an official course examiner

STEM Mentorship — Young Scientists Kenya (YSK)

Kenya | Jan 2020 – Dec 2021

Technical mentorship for secondary school student research teams participating in the Young Scientists Kenya national science and technology exhibitions. Provided engineering guidance across mechanical design, embedded systems, system architecture, prototyping, and technical report editing for automotive and aviation projects

Student Assistant - ASE301 Numerical Methods for Aerospace Engineers — Middle East Technical University

Turkey/ Northern Cyprus | Feb 2019 – Jun 2019

Appointed as Student Assistant for ASE301 (Numerical Methods for Aerospace Engineers) under Prof. Dr. Hakan Tarman based on highest academic grade merit (AA). Conducted consultation sessions and evaluated computational assignments covering ODE/PDE numerical solvers in MATLAB for a cohort of 30+ undergraduate students.

SI-PASS Leader & Instructor — MAT119 & MAT120 — Middle East Technical University

Turkey/Northern Cyprus | Sep 2018 – May 2019

Certified Supplemental Instruction (SI-PASS) Leader and Instructor at METU NCC, facilitating peer-assisted study sessions for undergraduate mathematics courses MAT119 (Calculus I: Calculus of Single Variable Functions) and MAT120 (Calculus II: Calculus of Multivariable Functions) throughout the 2018 Fall and 2019 Spring academic semesters.

LEADERSHIP & MENTORSHIP

Vice President, Aerospace Society — Middle East Technical University

Sep 2017 – Jun 2018

Executive leadership role co-managing student society operations, technical software workshops, faculty/guest seminars, and hands-on project teams. Spearheaded the society's Model Aircraft Project, facilitated MATLAB/Inventor technical courses, and established professional development channels for aerospace engineering undergraduates.

President, Model United Nations Society — Middle East Technical University

Sep 2017 – Jun 2018

Executive leadership role reviving and directing the dormant Model United Nations Society at METU NCC. Authored society charter/proposals, rebuilt an active member base of 30+ students, directed weekly parliamentary procedure and debate training workshops, hosted guest diplomacy seminars, and led a 17-member delegation to MUNTR 2018 in Antalya, Turkey

Secretary General, International Students Association — Middle East Technical University

Sep 2016 – Jan 2018

Executive leadership role serving as Secretary General of the International Students Association at METU NCC[cite: 3, 4, 25, 31]. Served as the highest-ranking African student representative on the executive board, directed day-to-day operations across 8 directorates, led recruitment interviews for student leadership positions, managed administrative approval pipelines, and ran point on major cultural, social, and sports events

LICENCES & CERTIFICATIONS

- Private Pilot Licence - Aeroplane (YK-9469-PL) | KCAA / Skylink Flight Services (2015)
- Flight Radio Telephony Operator's Licence (YK-9469-RL) | KCAA (2015)
- English Language Proficiency Rating (Level 5) | KCAA (2015)
- Remote Pilot Licence - Multirotor (YK-RPL-00013A) | KCAA / Kendrone Ltd (2021)
- Instructor Rating | KCAA / Kendrone Ltd (2021)
- EU A1/A3 Drone Pilot Licence | EASA (2023)
- Full Stack Software Development Certificate | eMobilis Technology Institute (2020)

LANGUAGES

English: Native / Fluent · **Swahili:** Native / Fluent · **German:** B2